3 A beam of unpolarised light of intensity $I_o$ is incident normally on a set-up consisting of three polarisers $P_1$, $P_2$ and $P_3$, arranged as shown in Figure 3.1.

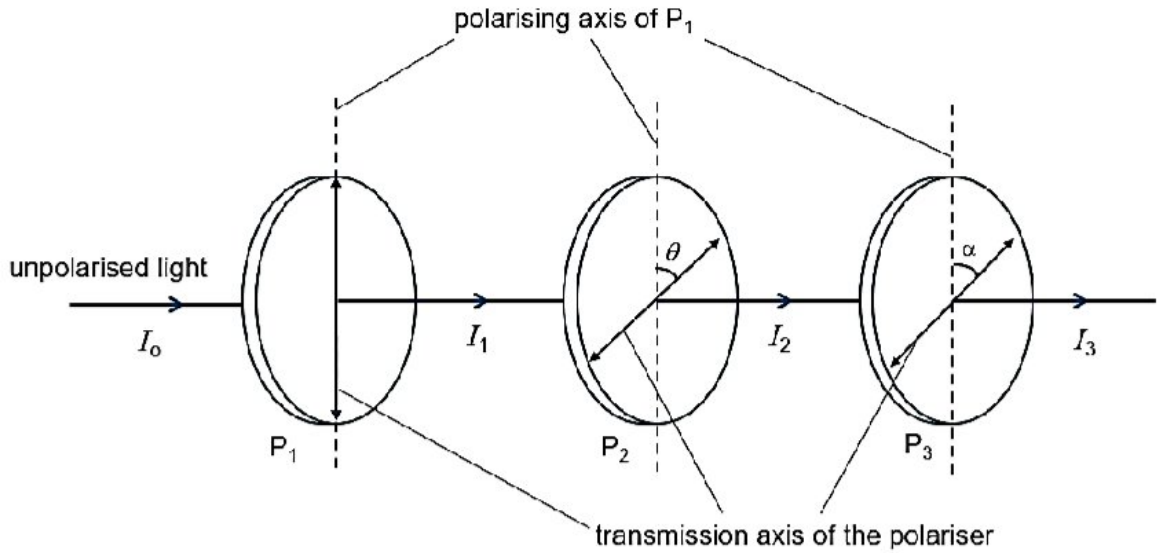



**Figure 3.1**

The light is plane polarised vertically after passing through $P_1$. The transmission axis of $P_2$ is at an angle $\theta$ to the vertical and the transmission axis of $P_3$ is at an angle $\alpha$ to the vertical.

The intensity of the light transmitted through $P_1$, $P_2$ and $P_3$ are $I_1$, $I_2$ and $I_3$ respectively.

**(a)** Calculate the transmitted intensity $I_3$ in terms of $I_o$ when $\theta = 25°$ and $\alpha = 90°$.

$I_3$ = ..... [2]

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**(b)** Angle $\alpha$ of polariser $P_3$ is kept fixed at 90° while angle $\theta$ of polariser $P_2$ is varied.

In Figure 3.2, sketch the variation of the amplitude of transmitted light after passing through $P_3$ with $\theta$. Numerical values on the vertical axis are not needed.

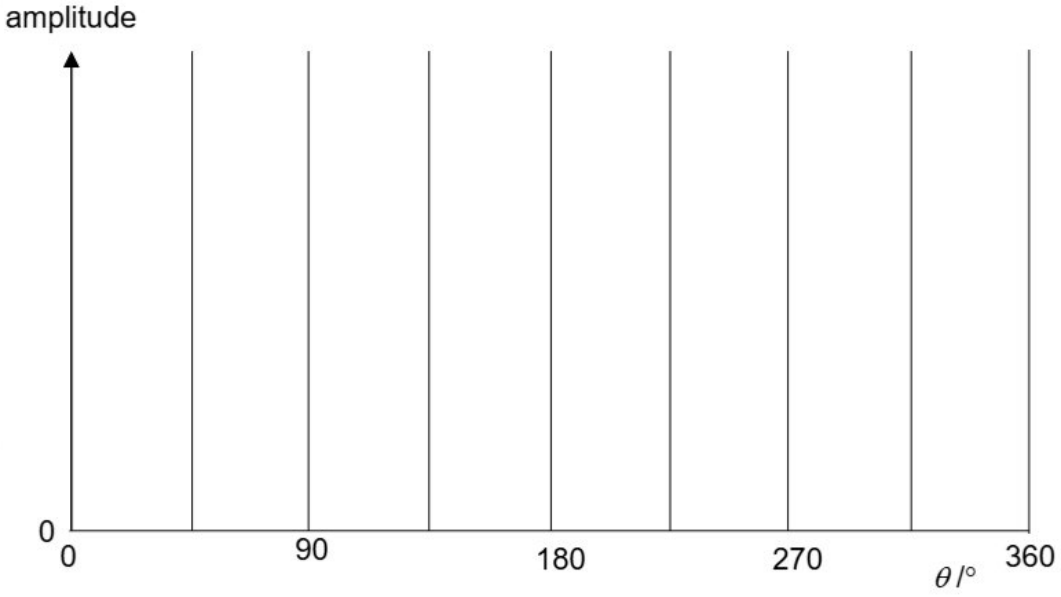



**Figure 3.2**

[2]

[Total: 4]